

TOBY CRUST

AI ENGINEER — LLM AGENTS • WORKFLOW AUTOMATION • INTERNAL TOOLING

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Software engineer building **LLM agents and workflow automation** that do real work for a team: an agent that turns Slack requests into tested pull requests, and a function-calling assistant that only changes a user's data after they approve it. Backed by three years of **full-stack** (Python, React, TypeScript) and **real-time** (Unity, C#) shipping for SaaS, government and nonprofit clients.

SELECTED WORK

SLIK • AI & INTERNAL TOOLING

Slack-to-VM AI Agent & Testing Pipeline

Built an internal AI automation pipeline integrated with Slack to speed up development and non-dev workflows. Non-technical team members specify requested project changes directly in Slack, which dispatches to an AI agent that provisions a virtual machine, applies code changes, executes test suites, and alerts me for final PR review and merge. Drastically cut routine work and accelerated overall team shipping velocity.

Python AI Agents Slack API VM Automation GitHub Actions Automated Testing CI/CD

TICKYTICK • PERSONAL PRODUCT

Function-calling assistant with human review

In-app AI assistant for TickyTick, my local-first task manager, built on Gemini function calling. The model never writes to the database: it calls one `propose_items` function, and the app validates the proposal, resolves names to real rows and shows it for review before anything is saved. Row ids never leave the device, and the prompt is ordered so the static rules and data snapshot reuse Gemini's implicit prefix cache, with only the timestamp changing per call. Served through a Supabase Edge Function with a usage quota.

TypeScript Gemini API Function Calling Prompt Caching Supabase Edge Functions Deno React

SLIK • NORTHERN TERRITORY GOVERNMENT

AR heritage app with conversational characters

Lead developer on a historical WWII AR app for the NT Government on iOS and Android, triggering on-site content via BLE beacons and geolocation, with Convai-powered conversational characters. Built with Unity Addressables to keep the download small and ship new content without a full release. Further locations are in production now, due to the success of the first.

Unity C# Convai AR Foundation BLE beacons Addressables

EXPERIENCE

Dec 2025 – present

Slik — Software Engineer, Sydney

- Built custom internal Slack-to-VM AI agent workflows to automate stakeholder requests, code modifications, and testing pipelines.
- Automated content and data pipelines with GitHub Actions and DigitalOcean Spaces.
- Lead developer on the NT Government AR app, from Unity build through to App Store Connect and Play Console release.

Oct 2024 – Oct 2025

Fabra — Software Engineer, Sydney

- Wrote internal tooling that cut manual steps out of the team's day-to-day backend operations.
- Took over a departing senior engineer's workstream and shipped share-design permissions and email validation end to end, front and back.
- Built features across Fabra's core SaaS product in React and TypeScript.

Feb – Oct 2024

Bardon Design — Junior Software Engineer, Bangkok

- Worked with the Louis Vuitton team on a gamified e-commerce experience built to lift engagement through interactive shopping.
- Built an interactive AR furniture selection app.

Summers, 2021 – 2023

Software internships — Auckland

- Built an AR Rube Goldberg machine level for GeoArGames, and a 2D arcade game embedded in the Yuri&Niels site.

SIDE PROJECTS

Holo Box — Slik R&D

Holographic display that tracks the viewer's head with machine learning and renders through an off-axis frustum, so a flat screen reads as real depth.

ML head tracking Unity Off-axis projection

Hexapod robot — robotics

Six-legged walking robot: Arduino firmware with inverse kinematics, driven over WiFi from a three.js leg editor through a Python bridge.

Python C++ three.js ESP8266

TOOLKIT

LANGUAGES

Python
TypeScript / JavaScript
C#
C++

AI & LLMS

AI Agents
Function Calling / Tool Use
Prompt Engineering
Gemini API
Convai

AUTOMATION

Slack API & Webhooks
VM Automation
GitHub Actions
Automated Testing
CI/CD

PLATFORMS

React
Supabase
REST APIs
Netlify / Vercel
DigitalOcean Spaces

EDUCATION

Bachelor of Design Innovation — Victoria University, 2021–2023

REFERENCES

Available on request — previous engineering leads at Fabra and Bardon Design are happy to speak.

OUTSIDE WORK

Robotics · 3D printing · Snowboarding · Waterpolo · Toastmasters